	Hence, apparent weight is smaller at equator.	A1
2ai	Gravitational force provides centripetal force	B1
	$\frac{GMm}{r^2} = \frac{mv^2}{r}$ and $v = \frac{2\pi r}{T}$ OR $\frac{GMm}{r^2} = mr\omega^2$ and $\omega = \frac{2\pi}{T}$	C1
	$r = \sqrt[3]{\frac{6.67\times10^{-11}\times6.0\times10^{24}\times(13.7\times24\times3600)^2}{4\pi^2}} = 2.4217\times10^8 \text{ m}$	A1
	$r \approx 2.4\times10^8 \text{ m}$	A0

2aii	Change in GPE $= -\frac{GMm}{r_{orbit}} - \left(-\frac{GMm}{r_{surface}}\right) = -GMm\left(\frac{1}{r_{orbit}} - \frac{1}{r_{surface}}\right)$ $= -6.67\times10^{-11}\times6.0\times10^{24}\times360\times\left(\frac{1}{2.4\times10^{8}} - \frac{1}{6.4\times10^{6}}\right)$ $= 2.2 \times 10^{10}$ J	C1 A1
2b	(Rock has minimum speed when all its KE is converted to GPE to reach infinity with zero GPE and zero KE) Loss in KE of rock = Gain in GPE or $\frac{1}{2}mv^2 = \frac{GM_{moon}m}{R_{moon}}$ $v_{min} = \sqrt{\frac{2GM_{moon}}{R_{moon}}} = \sqrt{\frac{2\times6.67\times10^{-11}\times7.4\times10^{22}}{1.7\times10^{3}\times10^{3}}} = 2.4\times10^{3}\ \text{m s}^{-1}$ Since $V_{rock} > V_{min}$, it will travel out into distant space **Alternative methods** Change in GPE from Moon surface to infinity $= \frac{GM_{moon}m}{R_{moon}} = \frac{6.67\times10^{-11}\times7.4\times10^{22}}{1.7\times10^{3}\times10^{3}}m = 2.9\times10^{6}\ m$ KE initial = ½ mv^2 = ½ x m x $(2.8 \times 10^3)^2$ = 3.9×10^6 m Since KE initial > change in GPE from surface to infinity, it will travel out into distant space.	M1 M1 A1 (M1) (M1) (A1)
3a	the *increase* in internal energy of a system is equal to the sum of the *heat supplied to*	A1